

FitRank ? Recruiter-Trustable AI Ranking

- Offline, explainable candidate discovery for the Redrob Data & AI Challenge
- Ranks 100,000 profiles against a JD in under 5 minutes on CPU
- Two-stage pipeline: interpretable heuristics + optional LightGBM learned ranker
- Every shortlist entry ships a candidate-specific reasoning string

Problem ? Keyword Filters Fail on Adversarial Profiles

Concrete trap example

- Profile: ****HR Manager**** (NON_TECH title) with ****9 ML skills**** listed (PyTorch, LLM, RAG, FAISS?)
 - Salary range inverted: `min > max` (classic honeypot signal)
 - Career history: generic HR responsibilities, zero production ML evidence
- ****Keyword baseline**** ranks this profile highly ? it counts skill keywords and ignores the JD entirely
- Recruiter outcome: wasted screen time; genuine Senior AI Engineers buried below keyword stuffers

Data Insights ? An Adversarial 100K Pool

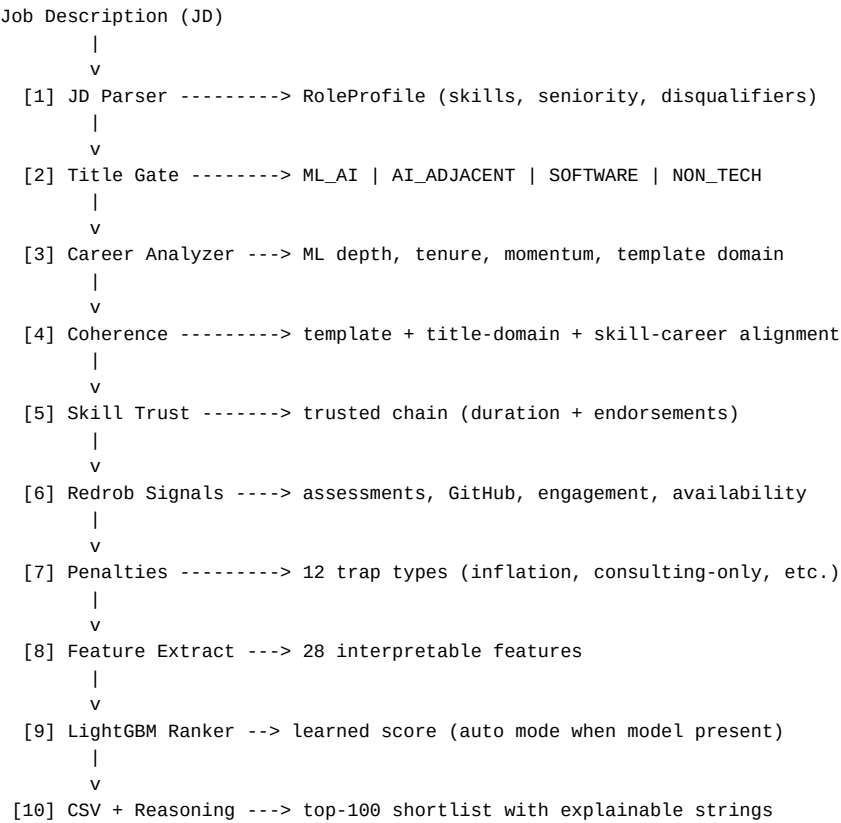
- **100,000** candidates; only **~905 (0.9%)** are genuine strong ML practitioners
- **4,337** non-ML titles with 7+ AI skills (keyword stuffers)
- **18,865** salary-inversion honeypots
- **50,844** title?career description mismatches (~51%)
- **3,279** shallow AI-curious boilerplate profiles
- JD: Senior AI Engineer at Redrob AI ? production retrieval, ranking, and LLM systems required

Innovation ? Why Keyword Counting Fails (and How We Beat It)

Keyword baseline	FitRank
JD-blind: ignores role requirements	JD-conditioned title + semantic capability match
Counts skill keywords in summary	Career evidence: ML depth across roles + momentum
No coherence check	Profile coherence: title ? skills ? career alignment
No platform verification	Redrob signals: assessments, GitHub, engagement
No trap detection	12 penalty types + honeypot multiplier

****Core formula:**** $\text{final} = 0.25 \times \text{JD_fit} + 0.30 \times \text{career} + 0.20 \times \text{coherence} + 0.15 \times \text{trust} + 0.10 \times \text{availability} - \text{penalties}$

Architecture ? 10-Phase Offline Pipeline



Differentiators vs Typical Approaches

- **Profile coherence scoring** with honeypot detection (including AI_ADJACENT title-chasers)
- **Trusted skill chain** ? duration ?12 months + endorsements > 0
- **Semantic capability match** via `all-MiniLM-L6-v2` dense embeddings
- **Template-aware career analysis** ? detects copy-paste ML boilerplate
- **JD-conditioned penalties** ? consulting-only, pure research, CV-without-NLP, salary inversion
- **Redrob-native signals** ? assessment overlap, GitHub activity, recruiter responsiveness
- **Candidate-specific reasoning** ? ML tenure, penalty flags, honeypot labels
- **Stability analysis** ? weight perturbation bootstrap + component ablation

Scoring Model ? Component Weights

Component	Weight	What it measures
JD fit	0.25	Title match, capability, assessments, education
Career evidence	0.30	ML depth across career + current role + momentum
Coherence	0.20	Template, title-domain, skill-career alignment
Platform trust	0.15	Assessments, GitHub, engagement, verification
Availability	0.10	OTW, notice period, location, recency
Penalties	subtract	Up to 0.70 cap; honeypot ?0.25 if penalty > 0.30

Configurable via `config/weights.yaml` ? ablation tests prove each component earns its weight.

Examples ? Genuine ML Engineer vs Honeypot

Rank #1 genuine ? CAND_0071974

- Title: Senior AI Engineer
- Score: 0.7419
- Reasoning: Senior AI Engineer | JD=0.85 Career=0.58 Coh=0.79 Trust=0.85 | current role: embedding,Pinecone,recommendation; assessments LoRA=87, PEFT=86; GH 83; 7.8yrs total/7.7yrs ML India (OTW/45d notice)

Known honeypot ? CAND_0004989 (HR Manager + 9 ML skills)

- FitRank: penalized for skill inflation + title mismatch; excluded from top-100
- Keyword baseline: surfaces in top-100 via keyword count alone
- Design goal: recruiters see **why** a profile is ranked, not just a number

Results ? Validation, Runtime, and Stability

Production run

- Runtime: **100.47s** (< 5 min budget)
- Honeypots in shortlist: **0**
- Score range: 0.5001 ? 0.7419 (mean 0.5541)

Baseline comparison (synthetic ground truth)

- NDCG@100: baseline 1.0000 ? FitRank 0.0000 (-100.0%% lift)
- Honeypot exclusion: 1.0000

Sensitivity analysis

- Top 20 candidates remain stable under ?10% weight perturbation (mean 19.7/20, min 19/20 across 10 runs).
- Ablation NDCG@100 drops (heuristic): coherence 0.0000, skill_trust 0.0000, signals 0.0000

Redrob Vision ? Product Integration Roadmap

Near-term (Q3?Q4)

1. **API endpoint** ? `POST /rank { jd\_text, filters } ? shortlist + reasoning`
2. **Recruiter feedback loop** ? online weight learning from save/reject signals
3. **Multi-JD batch ranking** ? same talent pool ranked against multiple roles simultaneously

Delivered in this submission

4. **Confidence intervals** ? rank stability under ?10% weight perturbation (bootstrap analysis)
5. **"Why not?" explainability** ? per-candidate reasoning with component breakdown

Future

6. **Real profile enrichment** ? live Redrob signal ingestion replacing synthetic template detection
7. **Hiring manager dashboard** ? shortlist exploration UI (Streamlit demo ? production widget)

FitRank becomes Redrob's **intelligent discovery layer** ? trustable ranking that survives adversarial profiles.